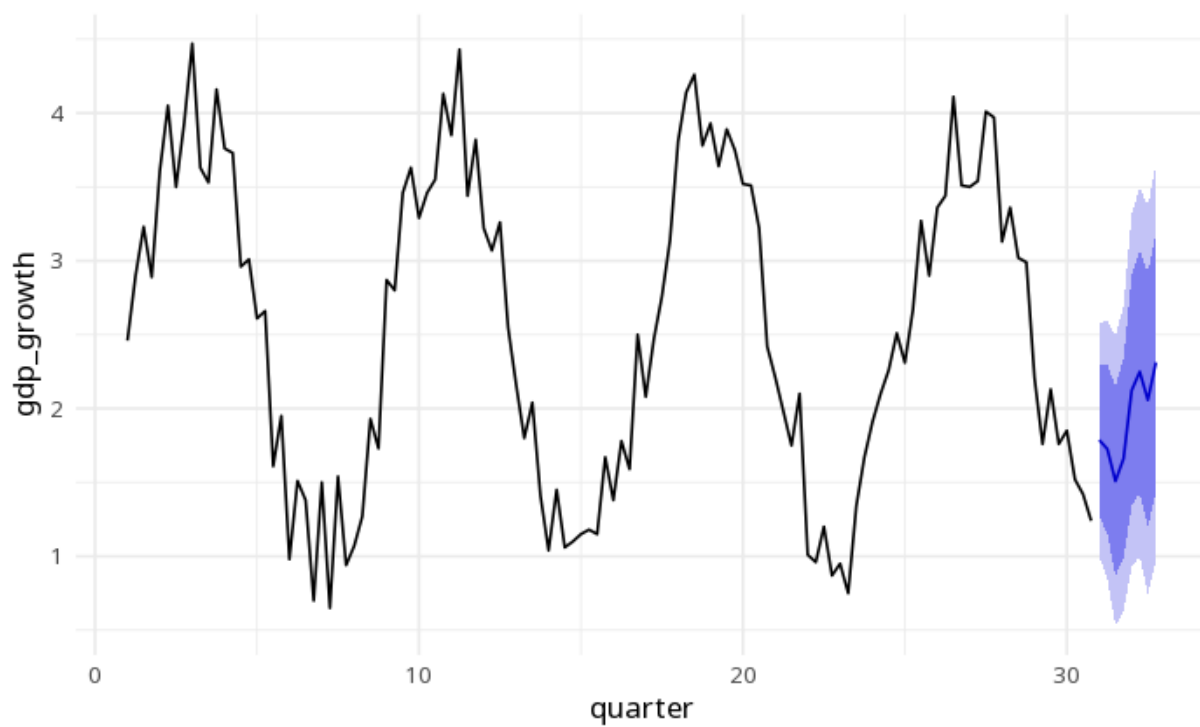
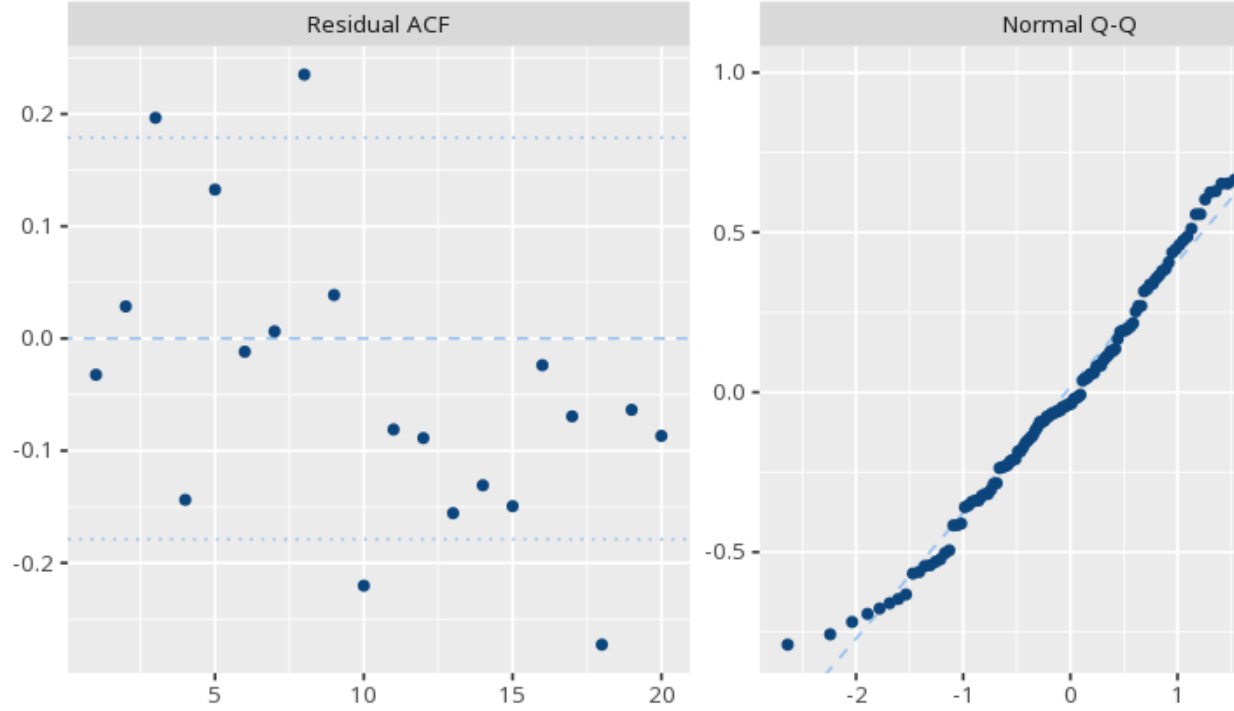


1 ARIMA / SARIMA

	(1)		
ar1	0.48 (0.11) [0.27, 0.68]		
ar2	0.33 (0.09) [0.16, 0.50]		
sar1	1.06 (0.12) [0.81, 1.30]		
sar2	−0.53 (0.12) [−0.76, −0.30]		
sma1	−0.70 (0.08) [−0.87, −0.54]		
intercept	2.49 (0.12) [2.25, 2.73]		
Num.Obs.	120		
σ^2	0.16		
Ljung–Box p	<.001		
AIC	133.95		
BIC	153.46		
Log.Lik.	−59.98		
Ljung–Box Q(5) = 23.82, p < .001 (lag 10, residual autocorrelation)			
Period	Forecast	80% PI	95% PI
1	1.79	[1.27, 2.30]	[1.00, 2.58]
2	1.73	[1.15, 2.30]	[0.85, 2.60]
3	1.51	[0.87, 2.15]	[0.53, 2.49]
4	1.66	[0.98, 2.34]	[0.63, 2.70]
5	2.12	[1.34, 2.90]	[0.93, 3.31]
6	2.25	[1.43, 3.07]	[1.00, 3.50]
7	2.06	[1.20, 2.91]	[0.75, 3.37]
8	2.31	[1.43, 3.19]	[0.97, 3.66]

Forecasts from ARIMA(2,0,0)(2,0,1)[4] with non-zero mean





The chosen $(p,d,q)(P,D,Q)$ orders describe the model. The Ljung–Box $Q(df)$ row tests whether the residuals still carry autocorrelation up to the reported lag (a large p means none is detected); use it with AIC/BIC to judge fit. The forecast table and plot give predictions with uncertainty bands. AIC/BIC compare models only when fit to the same series with identical differencing (d, D) and transformation; lower is better, but they are relative, not absolute, measures of fit. Method: R forecast package (Hyndman & Khandakar, 2008; Hyndman & Athanasopoulos, Forecasting: Principles and Practice).

An $ARIMA(2,0,0)(2,0,1[4])$ model was fit (AIC=133.95); the Ljung–Box test of residual autocorrelation gave $p < .001$.

Statistical basis: ARIMA/SARIMA model selection and forecasting (Hyndman R, Athanasopoulos G, Bergmeir C, et al. (2024). forecast: Forecasting functions for time series and linear models. R package.).